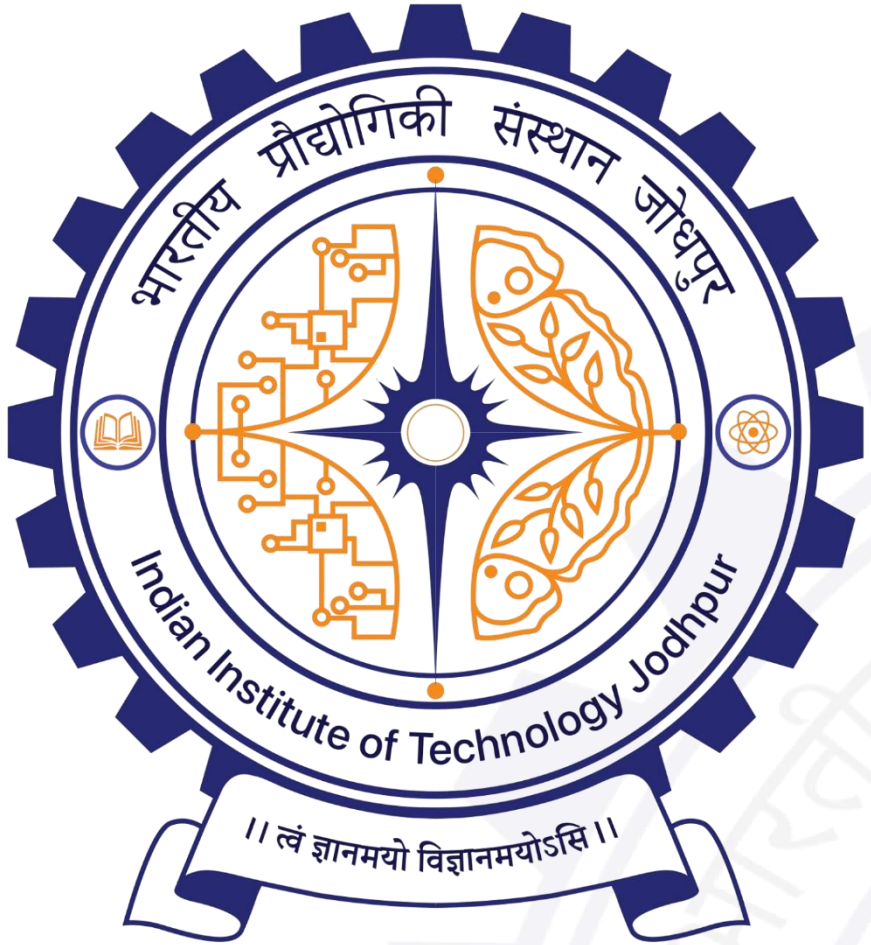




Sūryasiddhānta Computational Library with AI-Powered Query Interface

A Python library implementing classical Indian astronomical algorithms

MTP2

Ashutosh Nigam | M25AI2006

Supervisor: Dr Dinesh Mohan Joshi

Overview

Implement

Implement the computable methods from all fourteen chapters in a reusable Python library.

Explain

Show each calculation step and its source verses for a chosen date.

Evaluate

Compare results with DE440, stating the coordinate frame, time conventions and data sources.

Python Library: Computational Scope

Chapters	Implemented subject areas
1–3	Mean motion, true positions, place and time
4–6	Eclipse calculations and projected geometry
7–11	Conjunctions, stars, heliacal events, Moon and patas
12–14	Cosmography, instruments and reckoning of time

All interfaces use the same chapter modules for astronomical calculations.

Sūryasiddhānta: Key Algorithms

- Mean motion: elapsed days and revolution constants give mean planetary longitude.
- Manda and śīghra: epicycle corrections convert mean positions to corrected positions.
- Calendar calculations: use Sun and Moon longitudes to calculate tithi and related elements.
- Each chapter provides reusable Python functions for its calculations.

SS Corpus: Text Extraction

- Extracted Text from the English Burgess edition and the scanned Hindi edition – Vigyan Bhasya (Shri Mahaveer Prasad Shrivastava)
- Explored Gemma 4 E2B, Surya OCR 2, Sarvam Document Intelligence and Patram for OCR and translation.
- Completed 838-page Hindi OCR runs with Gemma and Surya OCR. A Gemma pass also translated all 838 pages into English.
- Saved page-level outputs, progress logs and reusable caches, with text, Markdown and Word exports.

SS Corpus: Chapter and Verse Exploration

- Organized extracted English text by chapter and verse, keeping verse text separate from commentary.
- Stored verse IDs, topics, source references and calculation-tool mappings in JSON/JSONL and SQLite.
- The app’s corrected Burgess export contains 500 numbered English verses across all 14 chapters.
- Verse Explorer, corpus search and chat retrieval use this export. Hindi OCR outputs remain separate from the bundled verse corpus.

System Architecture

Source texts → **Chapter functions** → **Structured results**

React + FastAPI

Calculation and book routes

Chat + tool registry

Same Python functions
for numerical answers

External MCP client

Same tool registry
through stdio transport

The Research Explorer reads saved DE440 comparison data separately from the library.

Technology Stack: Library and Interfaces

- Core library: Python, NumPy and Matplotlib.
- Backend: FastAPI, Pydantic and Uvicorn; REST APIs and MCP.
- Current web UI: React JS, TypeScript, Vite, React Router, HTML and CSS.
- 3D Visualization: Three.js, React Three Fiber and Drei.
- Equations and text: LaTeX, React Markdown.
- Earlier interfaces: Streamlit and Gradio.

Technology stack: AI, corpus and delivery

- AI: OpenRouter and DeepSeek APIs, with the OpenAI Python SDK and tool calling.
- Corpus: SQLite, JSON and JSONL; Markdown and Word exports.
- OCR and translation: Gemma 4 E2B, Surya OCR 2, Sarvam and Patram.
- Local model tools: LM Studio, Hugging Face Transformers and PyTorch.
- Document processing: PyMuPDF, OpenCV, Pillow and python-docx.
- Testing and delivery: pytest, Playwright, Docker, Docker Compose, Git and GitHub.

Current Status

- Python library: installable package with chapter modules and planet-wise imports.
- Interfaces: React and FastAPI show results, calculation steps and source links.
- AI query interface: chat and external MCP clients use the same tool registry.
- Learning paths: eight lessons cover all five classical planets, the Sun, Moon and Rahu.
- Exploration: chapter and verse browsing, a glossary and a 2D/3D Universe Explorer.

Implementation in numbers

56

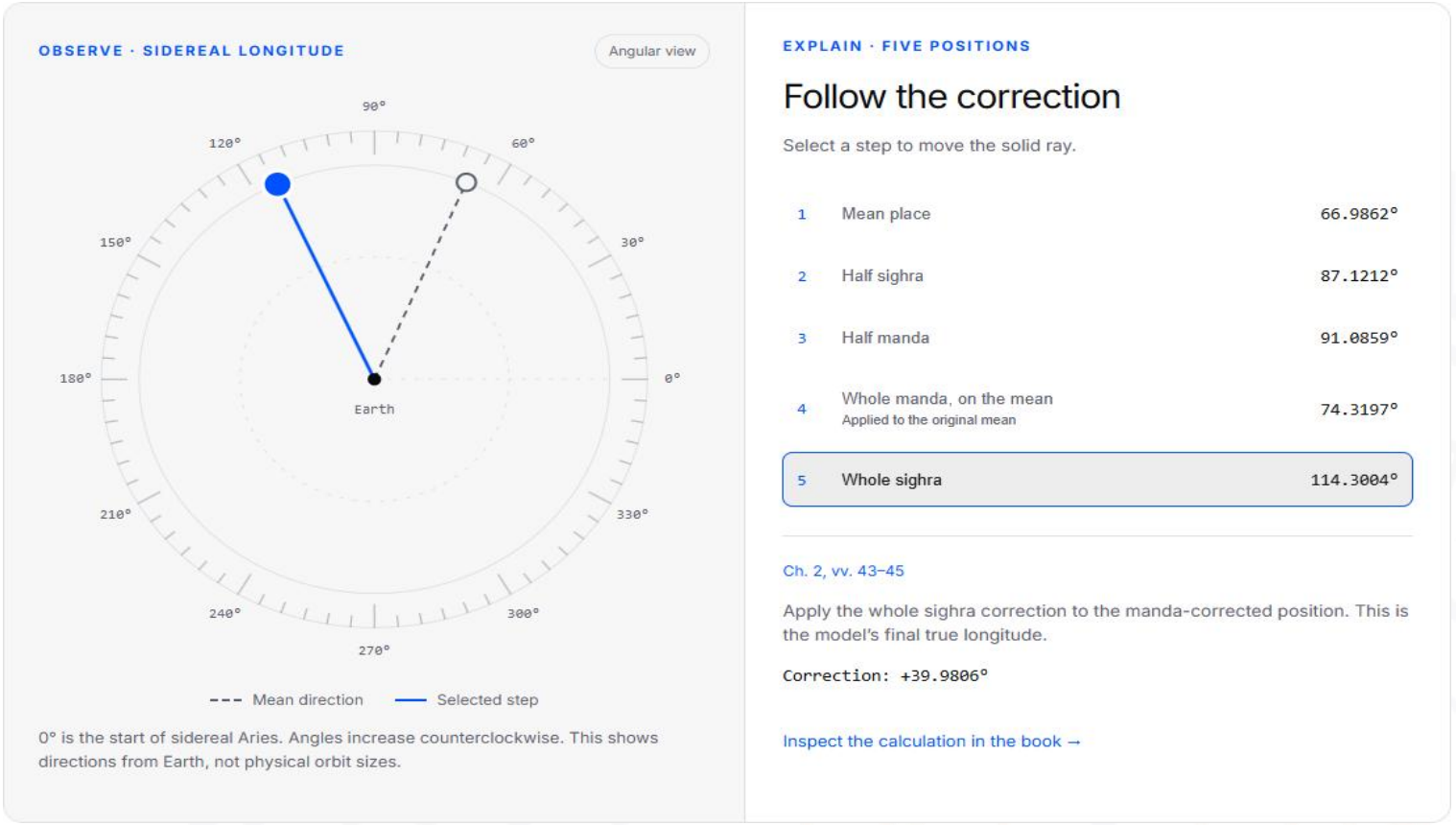
backend API endpoints

167

table placements in frontend code

Table placements are counted across the source code, not as tables visible on one screen.

User Interface: Planetary Learning Paths



- Eight paths: Sun, Moon, Mars, Mercury, Venus, Jupiter, Saturn and Rahu.
  Mars is shown as a worked example. Each lesson links to its source calculation.

Key Algorithms: Mars Worked Example

Stage	Sidereal longitude
Mean place	66.9862°
Half sighra	87.1212°
Half manda	91.0859°
Whole manda on the original mean	74.3197°
Whole sighra, final position	114.3004°

4 November 2026 · Ujjain mean time

Comparative Analysis: Research Explorer

OVERVIEW

Where does the error grow?

Time-aligned | Naïve time

Each cell shows RMS longitude error in degrees. Select a cell to inspect its evidence. Colour intensity uses one shared scale across this matrix.

Time-aligned · RMS error (degrees) · darker cells indicate larger errors

Body	1550 −1649	1650 −1749	1750 −1849	1850 −1949	1950 −2049	2050 −2149	2150 −2249	2250 −2349	2350 −2449	2450 −2549	2550 −2649
Sun	1.14	1.27	1.39	1.53	1.66	1.79	1.93	2.51	5.30	8.41	11.54
Moon	1.69	1.76	1.85	1.95	2.05	2.16	2.29	2.79	5.39	8.46	11.58
Mercury	3.31	3.25	3.20	3.16	3.14	3.13	3.14	3.45	5.78	8.71	11.75
Venus	4.16	4.74	5.38	5.96	6.61	7.23	7.77	8.55	10.17	12.58	15.14
Mars	1.88	2.21	2.67	2.96	3.36	3.85	4.16	4.95	7.98	11.20	14.56
Jupiter	2.09	2.24	2.45	2.87	3.33	3.84	4.29	4.19	2.12	1.94	4.45
Saturn	4.88	4.90	5.70	6.35	7.44	8.77	9.45	10.67	13.56	16.65	20.04
Rahu	1.50	1.79	2.10	2.42	2.76	3.10	3.43	3.43	1.63	2.16	4.56

Rahu is the ascending lunar node. The last windows include future extrapolations. These are sampled longitude errors, not eclipse timing errors.

Comparative Analysis: Time Conventions

Moon · 1550–1649 CE

Treatment	RMS error	Mean signed error
Naïve time	3.990°	-3.732°
Time-aligned	1.686°	-0.975°

The reference time changes. The historical SS equations stay the same.

The time-aligned comparison uses the Ujjain meridian conversion and an approximate Delta-T model.

Comparative Analysis: Long-Term Error

Body	1950–2049	2250–2349	2550–2649
Sun	1.658°	2.506°	11.536°
Moon	2.045°	2.791°	11.580°
Mars	3.356°	4.953°	14.559°
Venus	6.608°	8.555°	15.142°
Saturn	7.436°	10.671°	20.043°

RMS of tropical longitude residuals · future dates are model extrapolations

Project Contributions

A working, traceable computational edition

Users can follow a source rule through each calculation step and inspect the comparison results.

Next steps: study learning outcomes, add worked examples and check more algorithms against the text.

Limitations

Historical interpretation	The library implements computable methods and documents its assumptions and remaining gaps.
Numerical comparison	Ten-day sampling can miss peak errors. Longitude comparisons alone do not verify event timing.
Educational use	Lessons include understanding checks. Their effect on student learning has not been measured.

Future Work

- Check more algorithms against the source texts and document the remaining gaps.
- Add more worked examples and calculations to the eight learning paths.
- Run a structured student study to assess learning outcomes.
- State the time conventions and data sources used in each numerical comparison.

References and Project Links

GitHub: mrashutoshnigam/Surya_Siddhanta

Paper manuscript: Running the Text (computational edition)

Paper manuscript: SS algorithms vs JPL DE440, 1550-2649 CE

Project report: implementation, learning paths and evaluation

Source text: Ebenezer Burgess, English translation (1860)

SS corpus: OCR, translation and corpus pipeline (GitHub)



Thank You

Ashutosh Nigam (M25AI2006)

Supervisor: **Dr Dinesh Mohan Joshi**